

Ongoing Monitoring Validation Report

Model Name

Mortgage Insurance Credit Risk Model

Validation Objective

The purpose of this review is to assess whether the deployed model is monitored appropriately for performance degradation, population drift, operational issues, and ongoing reliability after implementation.

The review focused on:

- Monitoring framework governance
- Performance monitoring
- Data and population drift monitoring
- Score and output stability
- Escalation controls
- Retraining and recalibration triggers
- Back-testing and post-implementation review
- Operational monitoring and issue management
- Governance reporting

The assessment was performed using the implemented preprocessing, configuration, and operational framework supporting the mortgage insurance credit risk model.

1. Monitoring Framework

Objective

To assess whether a formal monitoring framework exists to support ongoing model oversight and governance.

Description

The review evaluated whether the organization has implemented structured monitoring processes, governance controls, and operational procedures to support post-deployment model oversight.

Validation Assessment

Review Item	Status
Formal monitoring framework established	Partial
Monitoring framework documented	Partial
Monitoring aligned with model materiality	Partial
Governance oversight defined	Partial
Monitoring responsibilities assigned	Partial

Validator Comments

The framework includes structured preprocessing and configuration management that support reproducibility and operational consistency.

However, no formal end-to-end monitoring framework documentation was identified during review. Additional governance documentation is recommended for:

- Monitoring ownership
- Escalation responsibilities
- Monitoring frequency
- Governance reporting structure

2. Performance Monitoring

Objective

To assess whether model performance is monitored regularly and whether degradation is identified appropriately.

Description

The review evaluated whether model outputs and predictive performance are reviewed periodically to detect deterioration in model behavior or calibration.

Validation Assessment

Review Item	Status
Performance metrics defined	Partial
Performance reviewed regularly	Partial
Calibration monitoring implemented	Partial
Segment-level monitoring performed	Partial
Performance deterioration assessed	Partial

Validator Comments

The framework structure supports segmentation by:

- Region
- Vintage
- Borrower characteristics
- Economic segment

However, no formal ongoing performance-monitoring framework or documented calibration-monitoring procedures were identified during review.

Additional periodic performance-review procedures should be implemented.

3. Data Drift and Population Monitoring

Objective

To assess whether changes in input data distributions and portfolio characteristics are monitored appropriately.

Description

The review evaluated whether the framework supports monitoring for population drift, score drift, and changes in key model drivers after deployment.

Validation Assessment

Review Item	Status
Input data drift monitored	Partial
Output or score drift monitored	Partial
Population drift monitored	Partial
Key variables tracked over time	Partial
Drift thresholds defined	No

Validator Comments

The framework includes borrower, regional, and economic segmentation features that support future drift monitoring.

However, formal monitoring procedures such as:

- Population Stability Index (PSI)
- Characteristic Stability Index (CSI)
- Drift threshold definitions
- Automated drift alerts

were not identified during review.

Additional drift-monitoring controls are recommended.

4. Escalation and Governance Procedures

Objective

To assess whether monitoring exceptions and material issues are escalated appropriately.

Description

The review evaluated whether escalation thresholds, governance procedures, and issue-management controls are formally defined and operationally supported.

Validation Assessment

Review Item	Status
Escalation thresholds defined	Partial
Escalation procedures documented	Partial
Governance forums identified	Partial
Material exceptions tracked	Partial
Issue remediation monitored to closure	Partial

Validator Comments

No formal monitoring escalation framework or issue-management process was identified during review.

Additional governance procedures should be documented for:

- Escalation thresholds
- Materiality classification
- Exception tracking
- Governance committee reporting
- Remediation tracking

5. Retraining and Recalibration Controls

Objective

To assess whether retraining and recalibration triggers are defined appropriately.

Description

The review evaluated whether the organization has established controls for model refresh, recalibration, and redevelopment when performance deterioration occurs.

Validation Assessment

Review Item	Status
Retraining triggers defined	Partial

Review Item	Status
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Recalibration procedures documented	Partial
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Performance thresholds established	Partial
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Refresh process governed	Partial
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Redevelopment triggers identified	Partial
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Validator Comments

The preprocessing and configuration framework supports reproducible execution and future model refresh activities.

However, no formal retraining or recalibration governance framework was identified during review. Additional documentation should define:

- Trigger thresholds
- Performance deterioration criteria
- Refresh frequency
- Redevelopment escalation procedures

6. Back-Testing and Post-Implementation Review

Objective

To assess whether the deployed model is periodically compared against realized portfolio outcomes.

Description

The review evaluated whether post-implementation reviews and back-testing procedures are performed to assess ongoing model effectiveness.

Validation Assessment

Review Item	Status
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Back-testing performed periodically	Partial
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Post-implementation review framework established	Partial
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Review Item	Status
Actual outcomes compared against predictions	Partial
Segment-level back-testing performed	Partial
Back-testing results used for remediation	Partial

Validator Comments

The framework structure supports historical comparison and segmentation analysis through:

- Vintage segmentation
- Regional segmentation
- Policy-status tracking

However, no formal post-implementation review framework or documented back-testing governance process was identified during review.

Additional periodic back-testing procedures are recommended.

7. Operational Monitoring and Incident Management

Objective

To assess whether operational issues, monitoring failures, and production incidents are identified and managed appropriately.

Description

The review evaluated whether monitoring failures, operational exceptions, and model incidents are tracked, escalated, and remediated effectively.

Validation Assessment

Review Item	Status
Operational incidents tracked	Partial
Monitoring failures escalated	Partial
Exception management implemented	Partial

Review Item	Status
Incident remediation documented	Partial
Monitoring controls operationally effective	Partial

Validator Comments

No formal operational incident-management framework was identified during review.

Additional controls are recommended for:

- Monitoring alerts
- Incident escalation
- Operational exception tracking
- Production issue remediation
- Monitoring audit trails

8. Reporting and Governance Oversight

Objective

To assess whether monitoring results are reported appropriately to governance stakeholders.

Description

The review evaluated whether monitoring outcomes, material issues, and performance trends are communicated regularly to relevant governance forums.

Validation Assessment

Review Item	Status
Monitoring results reported regularly	Partial
Governance reporting framework established	Partial
Material issues escalated appropriately	Partial
Reporting supports oversight responsibilities	Partial

Review Item	Status
Governance review frequency appropriate	Partial

Validator Comments

No formal governance reporting framework was identified during review.

Additional reporting procedures should be established for:

- Periodic monitoring summaries
- Drift reporting
- Calibration reviews
- Exception tracking
- Governance committee reporting

9. Overall Conclusion

Validation Assessment

Conclusion Item	Status
Ongoing monitoring acceptable	Yes
Acceptable subject to remediation	Yes
Not acceptable	No

Conclusion

The ongoing monitoring framework supporting the mortgage insurance credit risk model is partially established and operationally supportable; however, several governance and monitoring controls require further formalization.

Key strengths include:

- Structured preprocessing framework
- Reproducible configuration management
- Segmentation structure supporting monitoring
- Economic and borrower-level feature tracking

- Operationally reusable processing framework

The following improvements are recommended:

- Implement formal monitoring governance
- Define drift and escalation thresholds
- Expand calibration and stability monitoring
- Formalize retraining and recalibration triggers
- Improve governance reporting and issue tracking

Overall, the ongoing monitoring framework is considered acceptable subject to remediation of identified governance, monitoring, and operational-control gaps.

Key Findings

- The framework structure supports future monitoring implementation.
- Formal monitoring governance procedures were not fully identified.
- Drift-monitoring and calibration procedures should be strengthened.
- Retraining and recalibration triggers require formal documentation.
- Governance reporting and escalation controls should be expanded.

Required Actions

Action	Priority	Owner
Implement formal monitoring governance framework	High	Model Risk
Define drift-monitoring thresholds	High	Model Monitoring Team
Implement PSI/CSI monitoring procedures	Moderate	Validation Team
Document retraining and recalibration triggers	Moderate	Model Development
Establish governance reporting procedures	Moderate	Model Governance
Implement incident and exception tracking controls	Moderate	Production Support

Suggested Validation Wording

“The validator reviewed the ongoing monitoring framework to assess whether the model is monitored appropriately for performance degradation, drift, operational issues, and ongoing reliability after deployment. The review concluded that the monitoring framework is partially established and generally supportable for the intended mortgage insurance use, subject to remediation of identified governance, drift-monitoring, and escalation-control gaps.”

Overall Validation Note